HH-M-622D
May 18, 1977
SUPERSEDING
Fed. Spec. HH-M-622C
October 21, 1976

FEDERAL SPECIFICATION

MORTAR, REFRACTORY: HEAT-SETTING AND AIR-SETTING, BONDING (WET AND DRY TYPES)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers heat-setting and air-setting refractory mortars suitable for bonding refractory brick (see 6.1).

1.2 Classification.

1.2.1 Types, classes, and grades. Refractory mortars covered by this specification, shall be furnished in the following types, classes, and grades as specified (see 6.2):

Type I - Air setting - wet mortar.
Type II - Air setting - dry mortar.
Type III - Heat setting - dry mortar (low shrinkage).
Type IV - Heat setting - dry mortar.

Class 1 - Super duty.
Class 2 - High duty.
Class 3 - Medium duty.

Grade A - Commercial.
Grade B - Fine
Grade C - Extra Fine.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

HH-B-671 - Bricks, Refractory: Fire Clay.

FSC 9350

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:55Z
Check the source to verify that this is the current version before use.

UU-S-48 - Sacks, Shipping, Paper.
PPP-B-35 - Bags, Textile, Shipping, Burlap, Cotton and Waterproof Laminated.
PPP-D-705 - Drum: Metal Shipping, Steel, (Over 12 and Under 55 Gallon).

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletizing Unit Loads (40 in. x 48 in. 4-Way Partial and 4-Way Pallets).

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:55Z
Check the source to verify that this is the current version before use.

American Society for Testing and Materials (ASTM) Standards:

C24 - Test for Pyrometric Cone Equivalent (PCE) of Refractory Materials.
C92 - Test for Sieve Analysis and Water Content of Refractory Materials.
C113 - Test for Reheat Change of Refractory Brick.
C198 - Test for Cold Bonding Strength of Air-Setting Refractory Mortar (Wet Type).
C199 - Test for Refractoriness of Air-Setting Refractory Mortar (Wet Type).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P St., N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza,Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Types I and II. Mortar of types I and II shall be composed of pulverized fire clay, mullite, fused alumina, calcined fire clay, cyanite, diaspore, ball clay, and other substances that can be intimately mixed with silicate of soda or other material (excluding hydraulic cements), and which enable it to air-set when the compound is mixed with water.

3.1.2 Types III and IV. Mortar of types III and IV shall be composed of fire-clay materials basically similar to those used in the manufacture of fire-clay brick.

3.1.3 Particle size.

3.1.3.1 Grade A. Mortar of grade A fineness, when tested as specified in 4.4.1, shall have a fineness such that not more than 0.5 percent

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:55Z
Check the source to verify that this is the current version before use.

shall be retained on a United States standard sieve No. 16 and not more than 5 percent on a United States standard sieve No. 20.

3.1.3.2 Grade B. Mortar of grade B fineness, when tested as specified in 4.4.1, shall be such that not more than 0.5 percent shall be retained on a United States standard sieve No. 20 and not more than 5.0 percent on a United States standard sieve No. 30.

3.1.3.3 Grade C. Mortar of grade C fineness, when tested as specified in 4.4.1, shall be such that not more than 0.5 percent shall be retained on a United States standard sieve No. 20 and not more than 5.0 percent on a United States standard sieve No. 40.

3.2 Physical requirements.

3.2.1 Types I and II. Mortar of type I, classes 1, 2, and 3, when tested as specified in section 4, shall evidence physical characteristics that conform to the requirements of table I. Likewise, mortar of type II, classes 1, 2, and 3, when tested as specified in section 4, shall evidence physical characteristics that conform to the requirements of table I.

Table I. Physical requirements
for types I and II mortar, classes 1, 2, and 3

Property	Value
Refractoriness test temperature (minimum):	
Class 1	2910° F (1600°C)
Class 2	2730° F (1500°C)
Class 3	2550° F (1400°C)
Bonding strength, modulus of rupture, psi (minimum)	200
Particle size	1/
Water content (type I only): Percent (maximum)	25

1/Unless otherwise specified (see 6.2), the particle size of types I and II, in all classes shall conform to grade C (see 3.1.3.3).

3.2.2 Types III and IV. The physical requirements for type III, classes 1, 2, and 3, and type IV, classes 1, 2, and 3 mortar shall con-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:55Z
Check the source to verify that this is the current version before use.

form to the requirements of table II, when tested as specified in section 4.

Table II. Physical requirements for types III and IV mortar, classes 1, 2, and 3

Property	Value
Pyrometric cone equivalent (PCE), temperature	
Minimum	
Class 1	Cone No. 31 (1,683°C, 3061°F)
Class 2	Cone No. 27 (1,640°C, 2984°F)
Class 3	Cone No. 23 (1,605°C, 2921°F)
Bonding strength, Minimum, psi 1/	100
Particle size	2/
Linear shrinkage, (Type III only): Percent (maximum)	12

1/The test temperature shall be 1500°C for classes 1 and 2 mortar and 1350°C for class 3.

2/The particle size of types III and IV (all classes) shall be grades A, B, or C as specified (see 6.2).

3.3 Storage life. At any time within a 6-month period after Government storage, the mortar in a newly opened container shall not have stiffened or hardened to such an extent as to prevent its easy removal and mixing. The manufacturer shall certify to compliance with this requirement.

3.4 Workmanship. The refractory mortar shall be manufactured in accordance with commercial practice and the quality of workmanship shall be such as to produce mortar that is in accordance with the requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:55Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.2.1).
(b) Preparation for delivery (see 4.7).

4.2.1 Quality conformance inspection. Quality conformance inspection shall be performed on the sample lot selected in accordance with 4.3. This inspection shall include the examination of 4.5 and the tests of 4.6.

4.3 Inspection lot. For purposes of sampling, a lot shall consist of all refractory mortar manufactured as one batch and offered for delivery to the Government at one time.

4.4 Sampling. A random sample of filled containers shall be selecte from each lot in accordance with MIL-STD-105.

4.4.1 Sampling for examination. Examination of the containers of mortar shall be based on inspection level II and an Acceptable Quality Level (AQL) of 2.5 percent defective.

4.4.2 Sampling for tests. Tests of containers of mortar shall be based on inspection level S-3, and a AQL of 4.0 percent defective. The sample unit shall be 10 pounds of mortar. No more than one sample unit shall be drawn from any one container. The test sample shall be a 10 pound composite sample. Failure to meet any test requirements shall be cause for rejection.

4.5 Examination.

4.5.1 Inspection of filled containers. Each sample-filled container selected in accordance with 4.4.2, shall be examined for defects of the container and closure for evidence of leakage, and for unsatisfactory markings. Each sample-filled container shall also be weighed to determine the amount of the contents. Examination shall be conducted as specified in table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:55Z
Check the source to verify that this is the current version before use.

Table III. Classification of defects

Categories	Defects
Major 101	Container not type specified; damaged or evidence of leakage.
Minor 201	Marking, nomenclature and shipping data missing, incorrect, or illegible.

4.5.2 Lot acceptance tests. The test sample selected in accordance with 4.4.2 shall be subjected to the tests specified in 4.6.

4.6 Tests. Each sample selected in accordance with 4.4.2 shall be tested to determine compliance with this specification. Tests shall be conducted as specified in 4.6.1 through 4.6.7.

4.6.1 Particle size. The particle size of the mortar shall be determined in accordance with ASTM C92 by use of the procedure for wet sieve analysis.

4.6.2 Refractoriness. The refractoriness of types I and II mortar shall be determined in accordance with ASTM C199.

4.6.3 Pyrometric cone equivalent. The pyrometric cone equivalent for types III and IV mortar refractory shall be determined in accordance with ASTM C24.

4.6.4 Bonding strength for types I and II. The bonding strength of types I and II mortar shall be determined in accordance with ASTM C198.

4.6.5 Bonding strength for types III and IV. The bonding strength of types III and IV mortars shall be determined as specified in 4.6.5.1 through 4.6.5.3.

4.6.5.1 Mixing mortar. The portion of the sample to be used shall be thoroughly mixed with water to a troweling consistency for not less than 15 minutes.

4.6.5.2 Firebrick for bonding test. The firebrick shall meet the requirements for pyrometric cone equivalent given in HH-B-671.

4.6.5.3 Procedure for bonding test. The mortar for this test shall be of a good troweling consistency, mixed according to 4.6.5.1. The mortar shall be used to make five test joints, using brick specified in 4.6.5.2, according to ASTM C198 except that the test joints shall be placed in a kiln and subjected to the following firing schedule, in a neutral or oxidizing atmosphere before being tested for strength:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:55Z
Check the source to verify that this is the current version before use.

Classes 1 and 2 (1500°C) ASTM C113 schedule "F".
Class 3 (1350°C) ASTM C113 schedule "D".

After completion of the heating period, the test specimens shall be allowed to cool in the closed kiln for a period of at least 10 hours. When the specimens are cool, they shall be tested for modulus of rupture according to ASTM C198. Report the average of at least five determinations for modulus of rupture, of the mortar joint, as the bonding strength.

4.6.6 Shrinkage test, type III only. Three specimens shall be prepared using mortar mixed to a troweling consistency as in 4.6.5.2. The test specimens shall be prepared as follows:

(a) Place a metal frame that is 1/8-inch thick and forms a 1/2-inch border along all edges of the 9 by 4-1/2 inch face (see figure 1).
(b) Add a slight excess of mortar to the space inside the metal frame, and spread with a trowel to completely fill this space. Strike off excess mortar, using a straight edge. Do not trowel to a thick surface.
(c) Remove metal frame being careful to leave the applied coating undisturbed.
(d) Fill in the border with additional mortar and trowel to remove excess mortar, and make a feather edge along all edges of the 9 by 4-1/2 inch face. Make certain that the excess mortar is troweled to make a good seal at the feather edge. Be careful not to disturb the original coating. Allow the test specimen to air dry under normal laboratory atmospheric conditions for 24 hours, then place them in an oven and dry at 110°C ±5 percent for at least an additional 18 hours. The specimen shall then be removed from the drying oven and visually examined for cracks or opening up of the feather edge around the edge of the brick. The test specimens shall then be placed in a kiln, subjected to the heating schedules in a neutral or oxidizing atmosphere, and cooled as specified in 4.6.5.3. After the specimens have been removed from the kiln, and cooled to room temperature, they shall again be visually examined for cracks or opening up of the feather edge around the edge of the brick. The test specimens shall show no more than two hairline cracks and shall not pull away from the edge of the brick either after being dried or fired.

4.6.7 Water content. The water content of type I mortar shall be determined in accordance with the procedure described in section 4(a) of ASTM C92.

4.7 Preparation for delivery inspection. The packing and marking of the mortar shall be inspected to determine conformance with the applicable requirements of section 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:55Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Dry refractory mortar. The dry mortar shall be packed in bags conforming to the applicable requirements of PPP-B-35, or in sacks conforming to UU-S-48 as specified for level A packing.

5.1.1.2 Wet refractory mortar. The wet mortar shall be packed in drums conforming to type II of PPP-D-705.

5.1.2 Level B.

5.1.2.1 Wet refractory mortar. The wet mortar shall be packed as specified for level A.

5.1.2.2 Dry refractory mortar. The dry mortar shall be packed as specified for level A, except the sacks shall conform to UU-S-48 as specified for level B packing.

5.1.3 Level C. The refractory mortar shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules, or with National Motor Freight Classification rules.

5.2 Palletized unit load. When specified (see 6.2), shipping containers shall be palletized in accordance with MIL-STD-147.

5.3 Marking (see 6.2).

5.3.1 Civil agencies. Shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Types I and II. Refractory mortar, types I and II, covered by this specification, is particularly intended for bonding installations of firebrick in which temperatures may range approximately from that of the ordinary atmosphere to those encountered in the hottest zones of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:55Z
Check the source to verify that this is the current version before use.

chimney linings, indoor fireplaces, and other installations where the temperatures are too low to effect satisfactory bonding by ground fireclay or hot-set mortars but too high for hydraulic mortars. The mortar is not recommended for installations which are subjected to rain or dampness. Because of the rapid deterioration of the dry type of mortar when marketed in nonairtight containers, it is desirable to use such mortar shortly after manufacture except when it is stored in airtight containers.

6.1.2 Type III. Refractory mortar, type II, covered by this specification is particularly intended for bonding installations of fireclay brick subjected to prolonged service at high temperatures that necessitate tight joints and minimum gas- and air-leakage through walls.

6.1.3 Type IV. Refractory mortar, type IV, covered by this specification is intended for bonding firebrick subjected to high temperatures where control of air and gas leakage is not critical.

6.2 Ordering data. Purchasers should select the preferred options offered herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, and grade of mortar required (see 1.2.1, 6.3, table I, and table II).
(c) Particle size required for types I, II, III, and IV (see 3.2.1 and 3.2.2).
(d) Selection of applicable level of packing required (see 5.1).
(e) When shipping containers shall be palletized (see 5.2).
(f) Marking required (see 5.2).

6.3 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Armed Services Procurement Regulation (ASPR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 Classification changes. This document differs from the superseded specifications in respect to the following classifications:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:55Z
Check the source to verify that this is the current version before use.

New	Old
HH-M-622D	
Type I, class 1, grade C	Type I, grade A
Type I, class 2, grade C	Type I, grade B
Type I, class 3, grade C	Type I, grade C
Type II, class 1, grade C	Type II, grade A
Type II, class 2, grade C	Type II, grade B
Type II, class 3, grade C	Type II, grade C

New	Old
HH-M-622 D	HH-00611b(GSA-FSS)
Type I, class 1, grade C	Type I, class 3
Type I, class 2, grade C	Type I, class 2
Type I, class 3, grade C	Type I, class 1
Type II, class 1, grade C	Type II, class 3
Type II, class 2, grade C	Type II, class 2
Type II, class 3, grade C	Type II, class 1

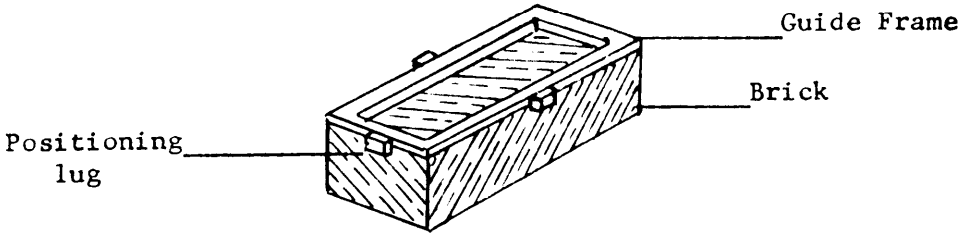
New	Old
HH-M-622 D	HH-M-622a
Type III, class 1, grade C	Grade A
Type III, class 2, grade C	Grade B

New	Old
HH-M-622 D	HH-M-622a
Type IV, class 1, grade A	Class C, grade 1
Type IV, class 2, grade A	Class C, grade 2
Type IV, class 3, grade A	Class C, grade 3
Type IV, class 1, grade B	Class F, grade 1
Type IV, class 2, grade B	Class F, grade 2
Type IV, class 3, grade B	Class F, grade 3

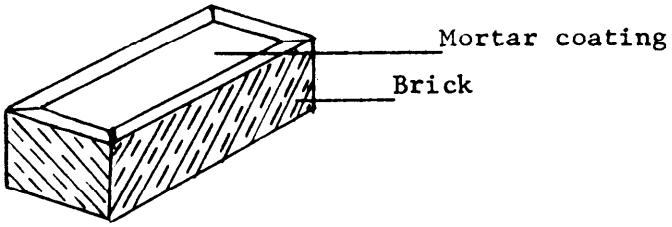
6.5 Supersession data. This specification includes the requirements of HH-C-451c, dated June 28, 1955; HH-M-611a, dated October 29, 1943; and HH-M-00611b (GSA-FSS), dated November 15, 1962.

6.6 International interest. Certain provisions of this specification are the subject of international standardization agreement (ABC). When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, international standardization channels including departmental standardization offices, is required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:55Z
Check the source to verify that this is the current version before use.



Brick with guide frame



Brick coated with mortar

Figure 1. Preparation of shrink test sample.

MILITARY INTEREST:

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activities:
Army - CE
DSA - GS

User activities:
Navy - MC, SH

Civil Agency Coordinating Activities:
GSA-FSS
COM-NBS

Preparing activity:
Navy - YD

Project No. 9350-0075

Orders for this publication are to be placed with General Services Administration, acting as agent for the Superintendent of Documents. See section 2 of this specification to obtain copies of documents referenced herein. Price 35 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:55Z
Check the source to verify that this is the current version before use.